



**SCHOOL OF COMPUTING AND ENGINEERING SCIENCES (SCES)
BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS
ENGINEERING (BSEEE)
EMBEDDED SYSTEMS**

LAB 2: PART A

COLLABORATORS:

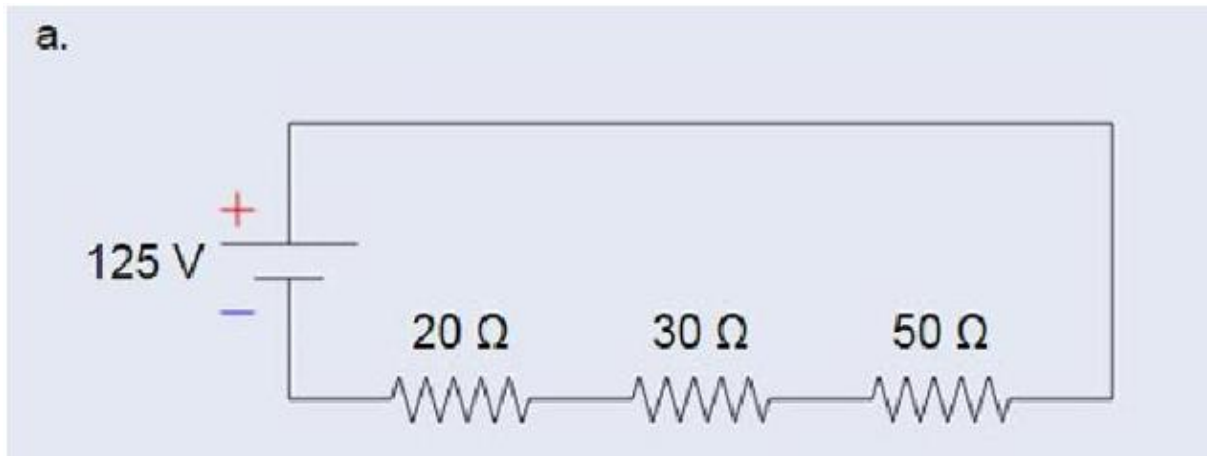
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Circuit (a): Series Circuit



Source voltage $V = 125 \text{ V}$. Resistors: $R_1 = 20 \text{ } \Omega$, $R_2 = 30 \text{ } \Omega$, $R_3 = 50 \text{ } \Omega$, connected in series.

1. Total Resistance

$$R_T = R_1 + R_2 + R_3 = 20 \text{ } \Omega + 30 \text{ } \Omega + 50 \text{ } \Omega = 100 \text{ } \Omega$$

2. Total Current

$$I_T = V / R_T = 125 \text{ V} / 100 \text{ } \Omega = 1.25 \text{ A}$$

In a series circuit the current is the same through every element, so $I_1 = I_2 = I_3 = 1.25 \text{ A}$.

3. Current, Voltage Drop, and Power per Resistor

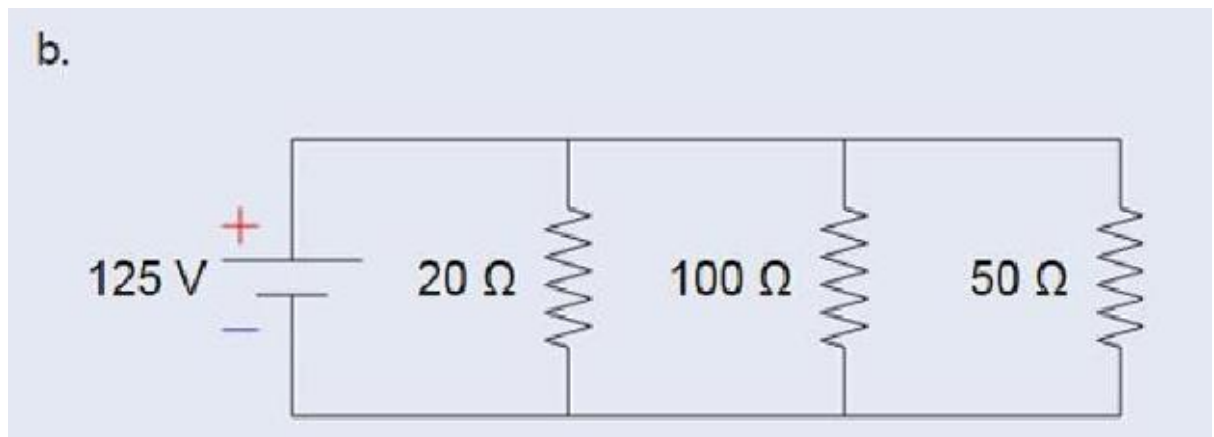
Resistor	Resistance / Ω	Current / A	Voltage Drop / V	Power / W
R1	20	1.25	25.0	31.25
R2	30	1.25	37.5	46.875
R3	50	1.25	62.5	78.125
Total	100	1.25	125.0	156.25

Voltage drop: $V_n = I \times R_n$. Power: $P_n = V_n \times I$

i. $V_1 + V_2 + V_3 = 25.0 \text{ V} + 37.5 \text{ V} + 62.5 \text{ V} = 125 \text{ V}$

ii. $P_T = V \times I_T = 125 \text{ V} \times 1.25 \text{ A} = 156.25 \text{ W}$

Circuit (b): Parallel Circuit



Source voltage $V = 125 \text{ V}$. Resistors: $R_1 = 20 \Omega$, $R_2 = 100 \Omega$, $R_3 = 50 \Omega$,

1. Total Resistance

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3 = 1/20 + 1/100 + 1/50 = 0.08\Omega^{-1} = 12.5\Omega$$

2. Total Current

$$I_T = V/R_T = 125\text{V}/12.5\Omega = 10\text{A}$$

3, 4, 5. Current, Voltage, and Power per Resistor

Resistor	Resistance / Ω	Voltage / V	Current / A	Power / W
R1	20	125.0	6.25	781.25
R2	100	125.0	1.25	156.25
R3	50	125.0	2.50	312.50
Total	12.5	125.0	10.00	1250.00

$$I_n = V / R_n. \text{ Power: } P_n = V \times I_n$$

$$I_1 + I_2 + I_3 = 6.25 \text{ A} + 1.25 \text{ A} + 2.50 \text{ A} = 10.00 \text{ A}$$

$$P_T = V \times I_T = 125 \text{ V} \times 10 \text{ A} = 1250 \text{ W}$$